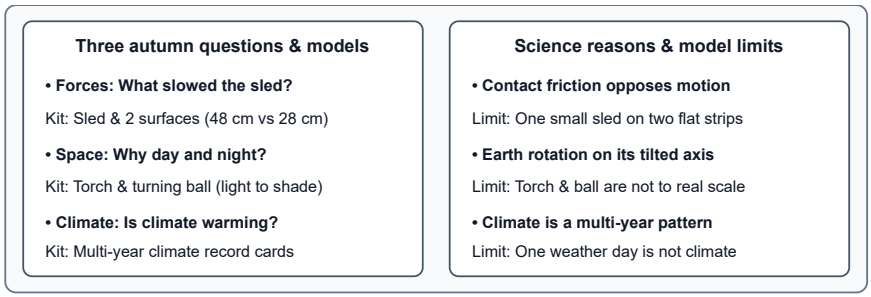
Reference sheets

Reference A · Autumn science recap kits, clues, and model limits

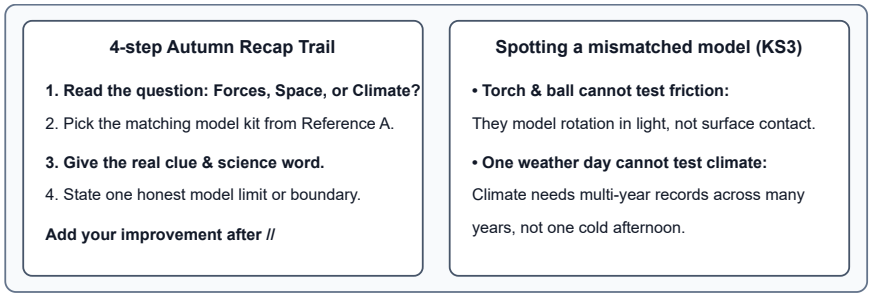


Autumn Recap Trail diagram. It links three questions to their models, clues, and limits.

Reference only. Do not write on this sheet. Match your question to one model kit below.

Science question.	Matching model kit.	Observed clue.	Key Stage 3 reason and limit.
1. Forces: What slowed the toy or sled?	Toy or sled and two marked surfaces (smooth vs rough mat).	Smooth rolled 48 cm; rough mat rolled 28 cm.	Reason: contact friction opposes motion. Limit: one small toy on two flat strips.
2. Space: Why does a place get day and night?	Torch and turning ball with a marked spot.	The marked spot turns from torch light into shade.	Reason: Earth rotates on its axis. Limit: torch and ball are not to real scale.
3. Climate: Is the climate warming over time?	Multi-year climate record cards across many years.	Long-term temperature pattern rises across decades.	Reason: climate is a multi-year pattern. Limit: one cold day is weather, not climate.
4. Honest model check.	Compare any simple classroom model with the real system.	Identify one detail the simple kit leaves out.	Reason: naming limits prevents over-claiming from a simple model.

Reference B · Four-step Recap Trail and mismatch repair



Four-step Autumn Recap Trail diagram. It shows model matching, clues, limits, and mismatch fixes.

Trail step.	What to do in your book or usual mode.	Example from today's recap.
1. Choose the model.	We used the ___ to answer why ___.	We used the sled and two marked surfaces to test stopping distance.
2. Give the clue & reason.	We observed ___ because ___.	Smooth gave 48 cm vs 28 cm on rough mat because contact friction slowed the sled.
3. State one limit.	One limit of this model is ___.	One limit is that we only tested one small sled on two flat indoor strips.

Made by Matt

Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use the arrival drawing. Match Boxes 1, 2, and 3 to contact friction, planetary rotation, and climate.

Help. Name the key scientific concept for each box.

2. Why does a mismatched model fail to give valid evidence for a scientific question?

Help. Explain why the model must match the variable.

3. In Box 1, smooth runs average 48 cm and rough runs average 28 cm. Explain this clue using contact friction.

Help. Compare the centimetre numbers and name the force.

4. State one limit of the surface desk test. State one limit of the globe model.

Help. Give one honest limit for each model.

You do tasks

1. Use Reference A and Reference B. Compare two autumn models. Explain how each model's clue supports its scientific conclusion. **Staff I ☐ S ☐ C ☐**

Help. Link Forces cm data to contact friction. Link the Space light clue to axial rotation.

2. Critique these two claims. Claim 1: 'One cold afternoon disproves climate warming.' Claim 2: 'A torch tests sled friction.' **Staff I ☐ S ☐ C ☐**

Help. Identify both mismatches using weather vs climate and model validity.

3. Evaluate one model's limits at Key Stage 3. Propose one improvement or boundary after //. **Staff I ☐ S ☐ C ☐**

Help. Explain what the simple model leaves out and how to bound the claim.

Optional participation-loop prompt. Share your chosen model and clue with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Why must a scientific model match the variable named in the question?

2. Compare the physical mechanism in the Forces kit with the mechanism in the Space kit.

3. State one limit of the two-surface sled test and one way to improve it.

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Aspire

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt